NAME:KRISHNAPRIYA P

REG NO:192511279

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CO2-AT2- Scenario Based Questions

Scenario 1:

Scenario

- A sender transmits 10 data frames (Frames 1–10) over a noisy wireless communication channel. Frame 4 is lost during transmission. Frame 7 is received with a bit error. Reliable communication is achieved using Automatic Repeat request (ARQ) protocols.

1. Stop-and-Wait ARQ – Reaction to Lost Frame 4

Working

- Sends only one frame at a time.
- Waits for an Acknowledgment (ACK) before sending the next frame.
- Uses a timer to detect lost frames.

When Frame 4 is Lost

- Frames 1, 2, and 3 are transmitted successfully.
- Sender transmits Frame 4.
- Frame 4 is lost due to channel noise.
- Receiver does not receive Frame 4 and sends no ACK.
- Sender waits until the timeout expires.
- Sender retransmits Frame 4.
- Receiver successfully receives Frame 4 and sends ACK 4.
- Transmission then continues with Frame 5.

2. Effect on Frames 5 and 6

- Stop-and-Wait allows only one outstanding frame.
- Frames 5 and 6 cannot be transmitted until Frame 4 is acknowledged.
- They remain in the sender's buffer.
- After ACK 4 is received:
    - Frame 5 is transmitted.
    - Frame 6 is transmitted.

- This results in higher transmission delay and lower channel utilization.

3. Go-Back-N ARQ

Working

- Sender can transmit multiple frames continuously without waiting for individual ACKs.
- Receiver accepts only in-order frames.

Handling Lost Frame 4

- Sender sends Frames 1–10.
- Frame 4 is lost.
- Receiver discards Frames 5–10 because they arrive out of order.
- Receiver repeatedly sends ACK for the last correctly received frame (Frame 3).
- After timeout, sender retransmits:
- Frame 4 → Frame 5 → Frame 6 → Frame 7 → Frame 8 → Frame 9 → Frame 10
- If Frame 7 contains a bit error, it and all following frames are retransmitted again.

Advantages

- Better throughput than Stop-and-Wait.
- Simple receiver design.

Disadvantages

- Unnecessary retransmissions reduce efficiency.

4. Selective Repeat ARQ

Working

- Sender transmits multiple frames.
- Receiver stores correctly received out-of-order frames.

Handling Lost Frame 4

- Frames 5, 6, 8, 9, and 10 are buffered.
- Only Frame 4 is retransmitted.
- Since Frame 7 has a bit error, only Frame 7 is retransmitted.
- Buffered frames are delivered once the missing frame arrives.

Advantages

- Higher bandwidth utilization.
- Fewer retransmissions.

Disadvantages

- Requires more memory and processing.

5. Comparison of ARQ Protocols

Go-Back-N: Retransmits Frame 4 and all following frames, discards out-of-order frames, medium efficiency.
Selective Repeat: Retransmits only lost/corrupted frames, buffers out-of-order frames, high efficiency.

Scenario 2: MAC Sublayer and Collisions

Scenario

Three office computers (A, B, and C) are connected to a single wired Ethernet hub, which means all computers share the same communication medium. Computers A and B begin transmitting data at exactly the same time, resulting in simultaneous transmission over the shared cable.

1. What network event happens when both signals meet on the wire?

Collision in Ethernet

When Computers A and B transmit simultaneously, their electrical signals travel through the same Ethernet cable. Since a hub broadcasts data to all connected devices, the signals interfere with each other, causing a collision.

Effects of Collision

- Both transmitted signals overlap and become corrupted.
- The receiver cannot distinguish the original data.
- The damaged frames are discarded.
- No acknowledgement is generated for the corrupted frames.
- Both computers must retransmit their data.
- Frequent collisions reduce network throughput and increase transmission delay.

Example:
If Computer A sends Frame A and Computer B sends Frame B at the same instant, both frames collide on the cable, making them unreadable.

2. How does CSMA/CD detect this event?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the access control protocol used in traditional Ethernet networks to detect and recover from collisions.

Working Process

1. **Carrier Sense** : Each computer first checks whether the communication channel is idle. If another device is transmitting, it waits until the channel becomes free.
2. **Multiple Access** :All computers connected to the hub share the same transmission medium. Any device can transmit when the channel is free.
3. **Collision Detection** While transmitting, each computer continuously monitors the cable. If the transmitted signal differs from the received signal, the device detects a collision.
4. **Jam Signal** : After detecting a collision, each transmitting computer sends a **Jam Signal**. This informs every device on the network that a collision has occurred.

5. **Stop Transmission** : Both devices immediately stop transmitting to avoid wasting bandwidth.

3. Why do A and B wait for a random time period before retransmitting?

Binary Exponential Backoff Algorithm : After a collision, Computers A and B do not retransmit immediately. Instead, they wait for a random backoff time calculated using the Binary Exponential Backoff Algorithm.

Reasons for Random Waiting

- Prevents another immediate collision.
- Gives every computer an equal opportunity to access the channel.
- Reduces network congestion.
- Improves overall network efficiency.
- Ensures fair access among all connected devices.

Working

- After the first collision, each computer selects a random waiting time.
- If another collision occurs, the waiting time range increases exponentially.
- This process continues until transmission is successful.

Advantages of CSMA/CD

- Detects collisions quickly.
- Reduces repeated collisions.
- Improves channel utilization.
- Ensures reliable communication over shared Ethernet.
- Provides fair access to all connected devices.

Scenario 3: Flow Control and Buffer Overflow

Scenario

A high-speed server sends a continuous stream of data frames to an older printer connected through a network. The printer processes data much more slowly and has a small memory buffer for temporarily storing incoming frames.

1. What problem happens if the server sends frames faster than the printer can clear its buffer?

Buffer Overflow: If the server transmits data faster than the printer can process it, the printer's memory buffer becomes full, resulting in buffer overflow.

Effects of Buffer Overflow

- Incoming frames cannot be stored.
- Excess frames are discarded.

- Data loss occurs.
- The sender must retransmit lost frames.
- Network bandwidth is wasted.
- Printing becomes slower due to repeated retransmissions.
- Overall communication efficiency decreases.

Example:
If the printer's buffer can store 5 frames but the server sends 15 frames continuously, only the first 5 frames are stored, while the remaining frames are dropped.

2. How does Sliding Window Flow Control stop the printer's buffer from overflowing?

Sliding Window Flow Control: Sliding Window Protocol controls the number of frames the sender is allowed to transmit before receiving acknowledgements.

Working Process

1. The printer advertises its available window size.
2. The server sends only the permitted number of frames.
3. The printer processes the received frames.
4. After processing, it sends an ACK indicating available buffer space.
5. The sender moves ("slides") the transmission window forward and continues sending.
6. If the printer's buffer becomes full, it advertises a Window Size = 0.
7. The server temporarily pauses transmission.
8. When buffer space becomes available, the printer advertises a new window size, and transmission resumes.

3. What role does the Window Size field in the acknowledgement frame play?

Window Size Field specifies the maximum number of additional frames that the receiver is capable of accepting.

Functions

- Controls the sender's transmission speed.
- Prevents receiver buffer overflow.
- Matches the transmission rate with the receiver's processing speed.
- Improves bandwidth utilization.
- Reduces unnecessary retransmissions.
- Ensures reliable and efficient communication.

Advantages of Sliding Window Flow Control

- Prevents data loss due to buffer overflow.

- Supports continuous and efficient transmission.
- Improves network throughput.
- Reduces retransmissions.
- Increases communication reliability.
- Makes better use of available bandwidth.

Scenario 4: Error Detection with CRC

Scenario

A device uses the Generator Polynomial whose binary representation is 1011 to generate a Cyclic Redundancy Check (CRC). The data bit string is 110101.

1. How many redundant bits (CRC bits) are appended to the data frame?

The number of CRC bits depends on the degree of the generator polynomial.

Given

- Generator Polynomial = **1011**
- Degree of polynomial = **3**
- Data = **110101**

Calculation

- Number of CRC bits = Degree of Generator Polynomial
- CRC bits = **3 bits**

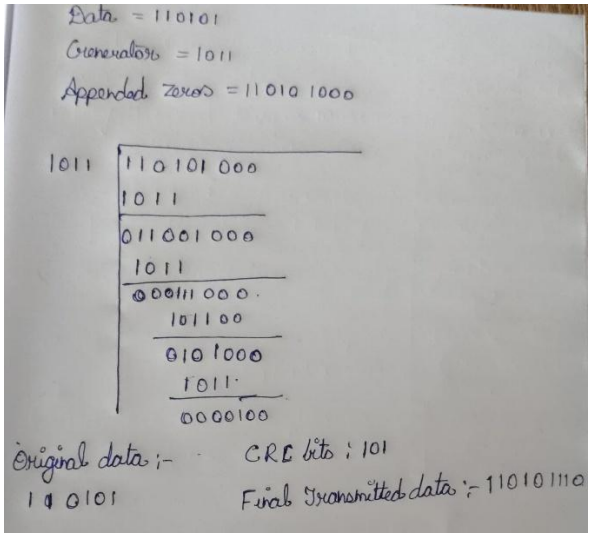
Data after appending zeros

Original Data : 110101

Append 3 zeros: 110101000

These three zeros are later replaced by the CRC remainder.

2. Modulo-2 Division to Find the CRC



3. How does the receiver detect a burst error?

Suppose the first two bits are flipped during transmission.

Original Frame

110101010

Received Frame

000101010

Receiver Operation

- The receiver divides the received frame by the same generator **1011**.
- If the remainder is **000**, the frame is accepted.
- If the remainder is **not 000**, an error is detected.
- Since two bits changed, the remainder becomes non-zero.
- The receiver concludes that the frame is corrupted.
- The damaged frame is discarded or retransmission is requested.

Advantages of CRC

- Detects single-bit and burst errors.
- Highly reliable error detection technique.
- Widely used in Ethernet, Wi-Fi, USB, and storage devices.
- Simple implementation using modulo-2 arithmetic.

Scenario 5: Framing and Byte Stuffing

Scenario

A communication link uses the special Flag Byte 0x7E to indicate the start and end of a frame. The payload itself also contains the byte 0x7E.

1. Why does the receiver think the frame ended early?

Problem

- The receiver treats every 0x7E as a frame delimiter.
- If 0x7E appears inside the payload, the receiver mistakes it for the end of the frame.
- The remaining data is interpreted as a new frame or discarded.
- This results in incorrect frame boundaries and corrupted communication.

2. How does Byte Stuffing solve this issue?

Byte Stuffing

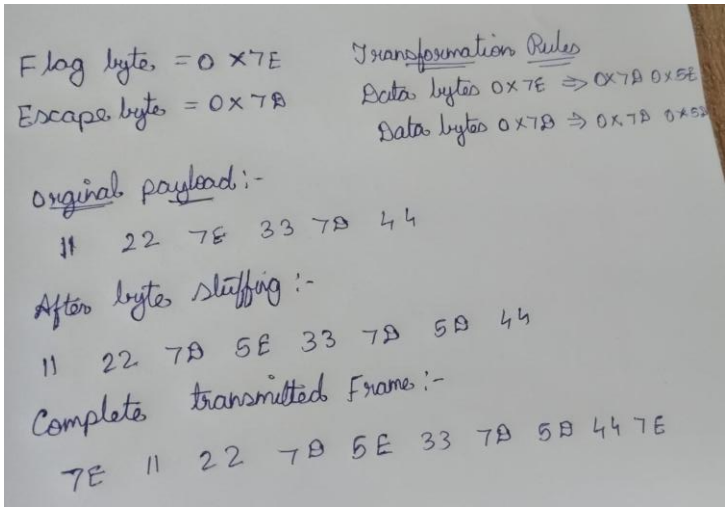
- The sender inserts an **Escape Byte (0x7D)** before every occurrence of the flag byte.
- The original flag byte is modified so it is no longer interpreted as a delimiter.

- At the receiver, the escape byte is removed and the original data is restored.

Working

- Detect special byte.
- Insert Escape Byte (0x7D).
- Modify the following byte.
- Receiver reverses the process.
- Original payload is recovered correctly.

3. Transformation of Special Bytes



Advantages of Byte Stuffing

- Prevents false frame detection.
- Preserves data integrity.
- Allows any byte value to appear inside the payload.
- Ensures correct frame synchronization.
- Widely used in protocols such as PPP (Point-to-Point Protocol).